ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

© 2026 TOBE ENERGY INC.

POLICY ANALYSIS

INVESTORS GUIDE TO 45V

Clean Hydrogen Tax Credit

Version: 2.0 · February 2026 **Classification:** Non-Confidential

EXECUTIVE SUMMARY

The Section 45V Clean Hydrogen Production Tax Credit (PTC), a cornerstone of the Inflation Reduction Act (IRA), offers a powerful, decade-long incentive of up to **$3.00 per kilogram**, fundamentally reshaping the economics of U.S. hydrogen projects. Final regulations from the U.S. Treasury and Department of Energy have now provided a clear and robust framework for implementation, de-risking key aspects of project development. However, a significant legislative challenge has emerged. A House-passed bill in mid-2025 proposes to accelerate the "begin construction" deadline for 45V eligibility from **December 31, 2027**, to **December 31, 2025**. While its passage in the Senate remains uncertain, this development has introduced a critical timing risk for the entire project pipeline.

KEY TAKEAWAYS FOR INVESTORS:

- **The Opportunity is Unprecedented**: The 45V credit (as a PTC or an elective **30% Investment Tax Credit**) makes the U.S. one of the most attractive markets globally for clean hydrogen investment.
- **Regulatory Clarity is Achieved**: Final IRS rules on emissions modeling (H_2-GREET), verification, and the use of Energy Attribute Certificates (EACs) provide a stable operating framework. The "GREET model safe harbor" is a crucial feature, locking in emissions pathways for a project's entire 10-year credit life.
- **A Critical Deadline Looms**: The proposal to sunset the credit for new projects starting after 2025 creates immense urgency. Developers and investors must now operate under the assumption that this accelerated deadline could become law.
- **Strategic Imperative is to "Safe Harbor"**: The primary risk mitigation strategy is to ensure projects "begin construction" before **December 31, 2025**, by either initiating significant physical work or, more commonly, by incurring at least **5% of the total project cost** under binding written contracts. Doing so would grandfather the project's eligibility for the 45V credit, irrespective of the legislative outcome.

1. THE 45V CREDIT: A GENERATIONAL INCENTIVE

Section 45V of the IRA provides a technology-neutral tax credit for the production of qualified clean hydrogen. Project owners can choose one of two incentive structures:

- **Production Tax Credit (PTC — §45V)**: A 10-year, per-kilogram credit based on the volume of hydrogen produced and its lifecycle carbon intensity. The maximum credit is **$3.00/kg** (inflation-adjusted) for projects meeting Prevailing Wage and Apprenticeship (PWA) labor standards.
- **Investment Tax Credit (ITC — §48)**: A one-time, upfront credit calculated as a percentage of the facility's eligible capital investment. The maximum credit is **30% of the project cost** for facilities designed to produce the cleanest hydrogen and meeting PWA standards.

The choice is irrevocable and strategic: the PTC favors projects with high production volumes and lower capital intensity, while the ITC provides significant upfront capital relief, benefiting capital-intensive projects and simplifying tax equity financing.

---

2. CREDIT TIERS AND ECONOMIC IMPACT

The credit value is tiered based on lifecycle greenhouse gas (GHG) emissions, measured on a "well-to-gate" basis using the DOE's 45VH2-GREET model. To qualify, a project's emissions must be below **4.0 kg CO_2e per kg of H_2**.

Table 1: Section 45V Credit Tiers (Assuming PWA Requirements are Met)

TIER	LIFECYCLE EMISSIONS (KG CO_2E PER KG H_2)	PTC (FULL RATE)	ITC (FULL RATE)
1	< 0.45 kg	**$3.00/kg**	**30%**
2	0.45 – 1.5 kg	**$1.00/kg**	**10%**
3	1.5 – 2.5 kg	**$0.75/kg**	**7.5%**
4	2.5 – 4.0 kg	**$0.60/kg**	**6%**

> **NOTE:** Base rates are 20% of the full rate if PWA standards are not met. All PTC values are indexed to inflation. The 5x multiplier for meeting PWA requirements makes compliance a near-universal goal for economically viable projects.

---

3. THE CRITICAL DEADLINE: NAVIGATING LEGISLATIVE HEADWINDS AND THE 2025 SAFE HARBOR

While current law provides a generous runway, recent legislative action has become the single most important variable for near-term project planning.

- **Current Law**: A qualifying facility must begin construction by **December 31, 2027**.
- **Proposed Legislation (House-Passed Bill)**: This deadline would be accelerated to **December 31, 2025**. The Senate's position is the pivotal factor, with significant industry lobbying to preserve the original timeline. However, investors cannot afford to wait for political resolution. Prudent strategy dictates action before year-end 2025.

The Safe Harbor Solution:

To mitigate this legislative risk, projects can be grandfathered under current law by establishing that they have "begun construction." The IRS provides two well-established methods:

1. **PHYSICAL WORK TEST**: Initiating physical work of a significant nature, either on-site (e.g., pouring foundations) or off-site for custom-ordered project components under a binding contract.
2. **5% Safe Harbor**: Incurring at least **5% of the total project's capital cost**. This is the more common path for large projects and can be satisfied through milestone payments for long-lead equipment (e.g., electrolyzers, reformers) under binding written agreements.

Once construction has begun, projects are typically afforded a four-year window (e.g., by Dec. 31, 2029 for a 2025 start) to be placed in service to satisfy continuity requirements automatically.

---

4. THE REGULATORY RULEBOOK: KEY PROVISIONS FROM TREASURY AND DOE

Finalized regulations provide the operational certainty needed for project financing and execution. Key provisions include:

- **Emissions Modeling**: Use of the 45VH2-GREET model is mandatory. Critically, a "beginning of construction emissions safe harbor" allows a project to lock in the GREET model version in effect when it starts construction for its entire 10-year credit period, protecting it from adverse future updates.
- **Sourcing Clean Electricity**: For electrolytic hydrogen, producers can use Energy Attribute Certificates (EACs) to document the use of clean electricity. The rules require EACs to be from the same grid region

and matched temporally. The temporal matching requirement is **annual through 2029**, transitioning to **hourly matching from 2030 onward**, giving the industry a runway to adapt.

- **Verification**: Annual third-party verification reports are required to substantiate production volumes and emissions intensity. This report is a non-negotiable attachment to tax filings for claiming the credit, making operational rigor paramount.

---

5. INVESTOR IMPLICATIONS & STRATEGIC OUTLOOK

The convergence of regulatory clarity and legislative risk creates a unique market dynamic.

- **Urgency & De-risking**: The primary focus for all stakeholders in the U.S. hydrogen market through H2 2025 should be on activities that satisfy the "begin construction" safe harbor. This includes finalizing EPC contracts, ordering long-lead equipment, and securing financing to make initial payments.
- **Diligence Focus**: Investors and lenders must now add "safe harbor qualification" as a key diligence item, scrutinizing contracts and payment schedules to ensure they meet the **5% threshold** by year-end.
- **Market Bifurcation**: A successful acceleration of the deadline would create a bifurcated market: a closed set of highly valuable, grandfathered projects eligible for 45V, and a second wave of projects that would be uneconomic without a new subsidy regime. This places an immense premium on projects that successfully safe-harbor in 2025.

While the legislative outcome is pending, the strategic path forward is clear. By leveraging the IRS safe harbor provisions, developers and their capital partners can secure access to the transformative 45V credit and insulate their projects from political uncertainty.

© 2026 Tobe Energy Corp. | tobe.energy